



A review for image–text matching from deep learning perspective

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ABSTRACT

The image–text matching task involves retrieving images guided by the given text descriptions or finding text descriptions using image queries. It guarantees various downstream tasks, including cross-modal retrieval, image captioning, text-to-image synthesis, and multi-modal neural machine translation. It can also empower several real applications, such as multi-modal search engines and vision-language AI assistants. Existing works try to resolve this task based on feature enhancement-based, cross-modal alignment-based, and vision-language pretraining-based approaches according to different design perspectives. Specifically, feature enhancement-based approaches usually elaborate on designing model architectures, which aim to reinforce the semantic representation abilities of the visual and textual features. Besides feature enhancement, how to make images interact with texts is equally important. To this end, the cross-modal alignment-based approaches are proposed to establish the interactions between the image regions and textual words for the strengthening of the discrimination ability on the matched region-word pairs. Moreover, to further improve the retrieval accuracy to accelerate the model implementation and application in a real scenario, vision-language pretraining-based methods are designed to reduce the gap between image and text in a fine-tuning manner or with the help of multimodal large language models. Therefore, the performances of vision-language pretraining surpass those following the other two types of methods, and the capabilities of cross-modal alignment-based approaches are the runner-up. In this work, existing related methods for the image–text matching task are first systematically sorted and analyzed. Later, the details of each method are categorized and concluded, consisting of motivations, advantages, and disadvantages. Then, the performance comparisons are observed and analyzed qualitatively and quantitatively. Finally, we prospectively summarize the possible unexplored issues and potential future solutions.

1. Introduction

Image–text matching aims to retrieve images from given text descriptions or find text descriptions by image guidance. This task is guaranteed for various downstream tasks, such as cross-modal retrieval [1,2], image captioning [3,4], text-to-image synthesis [5,6], multi-modal neural machine translation [7,8], etc. It is also the assurance of several real applications, such as multi-modal search engines and vision-language AI assistants. The core of this task is to eliminate the heterogeneous gap between image and text modalities. Current mainstream works attempt to resolve this task based on three significantly different directions, including Feature Enhancement-based (FE-based), Cross-Modal Alignment-based (CMA-based), and Vision-Language Pretraining-based (VLP-based).

The FE-based direction is in an attempt to enhance the semantic representation abilities of the visual or textual features. To this end, three kinds of strategies are usually considered for visual clues excavation and textual rules learning. At first, some researchers [9–18]

devote to reinforcing the image or language features solely with the help of multi-regional multi-label convolutional neural network [9], position prediction [11], relationship reasoning [13], syntactic structure understanding [17], and so on. Although impressive performances are achieved step-by-step, merely enhancing the single-modality feature may trap the model into the dilemma of cross-modal semantic misalignment. Sequentially, for the remedy of the cross-modal semantic misalignment, existing works shift attention toward jointly reinforcing the image and text features to align cross-modal semantic information [19–24]. In general, they simultaneously establish region-dependent relationships and explore meaningful words on the image and text modalities, respectively. Besides, there also exists a strategy that tries to design novel learning schemes [25–36] to mine the latent information hidden in the image and text modalities, e.g., designing reasonable loss functions, constructing powerful fusion models, and proposing targeted attention mechanisms.

In addition to feature enhancement, exploring varying interactions between image regions and text words is equally important, which can

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1566-2535/© 2025 Elsevier B.V. All rights reserved, including those for text and data mining, AI training, and similar technologies.

strengthen the discrimination ability of the aligned region-word pairs. Present cross-modal alignment-based (CMA-based) can be roughly classified into attention mechanism [37–46], progressive learning mechanism [47–54], and reasoning mechanism [55–62] three categories in terms of the differences of learning targets. The methods based on the attention mechanism typically focus on cross-attention boosting, bidirectional attention introduction, redundant attention elimination, etc. However, most of them neglect to recurrently use the learned features, leading to the successive disappearance of useful information. To resolve this challenge, other works tend to resort to the progressive learning of useful information in a multi-step manner. To be specific, the approaches typically attend to the recurrent memory for iterative matching [47,50], dynamic routing for optimal path decision [48], and coarse-to-fine context capturing [51]. In general, the results following progressive learning mechanisms are superior to those of attention mechanisms, while they are trapped in the imbalance between efficiency and effectiveness. To break the dilemma caused by progressive learning mechanisms, the techniques applying reasoning mechanisms are proposed to excavate the latent relationships no matter hidden in the text and image modalities. The latent relationships are generally excavated with the assistance of the object graph, scene graph, knowledge graph, etc. By excavating latent relationships, the proposed models can match the image–text pairs with complex matching patterns.

Although FE-based and CMA-based works have obtained impressive performances gradually, they are still difficult to deploy in real scenarios because of their unsatisfactory retrieval accuracy. For this aim, several researchers take the prevalent VLP-based schemes into account to either fine-tune specific image–text matching models or resort to the impressive abilities of multimodal large language models, whose abilities are superior to those of models trained from scratch. The expectations of VLP-based schemes either enrich the multiple modalities with the help of the imagination ability [63] of the large models or reduce the inherent gap inter-modalities drawing inspiration from the learning abilities [64,65] of the large models.

Although the task of image–text matching has extremely high application value and thus has attracted increasing attention, the related reviews are relatively fewer than the other interdisciplinary tasks [66,67] involving computer vision and natural language processing. Therefore, it motivates us to sort out and review related literature to stimulate the interest of the readers, make the researchers thoroughly explore this field, and accelerate the facilitation of the application of this task. To the best of our knowledge, it is the first research review on the task of image–text matching. The structure of this survey is depicted in Fig. 1. The main contributions of this work are summarized as follows: (1) The methods of image–text matching are first systematically and comprehensively reviewed; (2) The solutions of each method, including motivations, advantages, and disadvantages are illustrated in a categorized manner; (3) The performance comparisons are observed and analyzed qualitatively and quantitatively; (4) The possible unresolved challenges and potential future solutions are prospectively summarized.

2. Method analysis

The image–text matching task aims to retrieve images based on the guidance of the textual descriptions or find language descriptions with the help of the image queries. As the inherent gaps between image and text modalities, image–text matching is a significantly challenging task. To this aim, existing works can be roughly classed into three types: including Feature Enhancement-based (FE-based), Cross-Modal Alignment-based (CMA-based), and Vision-Language Pretraining-based (VLP-based). The timeline for the three types of methods is as depicted in Fig. 2. Besides, a summary of methods for the image–text matching task can be depicted in Fig. 3. Furthermore, the simplifying method structures are shown in Fig. 4. The brief introduction about different kinds of methods are demonstrated in Table 1, Table 2, and Table 3.

2.1. FE-based methods for image–text matching

FE-based methods are expected to reinforce the semantic understanding of the image and text features, enhancing the feature representation abilities of the visual and textual features. According to how to excavate or learn the visual clues or textual rules, current FE-based methods can be further divided into three directions, i.e., *single-modal semantic reinforcement*, *multimodal semantic reinforcement*, and *novel learning scheme*.

2.1.1. Single-modal semantic reinforcement

The works [9–18] located in this direction are devoted to reinforcing the image or language features solely. To achieve this, Huang et al. take the lead to propose the SCO [9] to learn semantic concepts and order to improve the image representation, which can address the challenge of large visual semantic discrepancy raised by the shortage of high-level visual semantic information. SCO predicts the semantic concepts (e.g., objects, properties, and actions) of images using a multi-regional multi-label CNN, designs a context-gated sentence generation scheme for semantic order learning as different orders of semantic concepts have different meanings, and introduces a conventional LSTM to learn the sentence representation. Although SCO reinforces the representation of the visual features, the salient features hidden in them are almost ignored and lead to limited retrieval results. As a remedy, subsequent works can be divided into **Salient Feature Extraction**, **Relation Reasoning**, and **Rule Excavation** three categories.

Salient Feature Extraction. Wang et al. focus on the position of the objects and propose a Position Focused Attention Network, PFAN [11], to investigate the relation between the visual and textual views. The architecture of PFAN is depicted in Fig. 5. Specifically, PFAN first decomposes the given image into blocks for the assurance that the relative position of regions can be inferred. Then, an attention mechanism is proposed to reason the relationships between blocks to generate valuable position information, enabling the enhancement and establishment of region expression and reliable relationships across modalities. Meanwhile, encountering the inherent semantic gap between image and language, Ji et al. propose a Saliency-guided Attention Network (SAN [12]) characterized by building an asymmetrical link between image and language to efficiently learn a fine-grained cross-modal correlation. SAN mainly includes three parts, i.e., a saliency detector, a saliency-weighted visual attention, and a saliency-guided textual attention, which are individually responsible for providing the visual saliency information, learning more discriminative visual features, and capturing powerful textual representations that are synchronized with visual clues. Despite the methods extracting salient features above having significance and value, they are confronted with how to align images to multiple textual descriptions with semantic diversity. With this in mind, Qu et al. present a Context-Aware Multi-view summarization network, CAMERA [16] for short, to summarize context-enhanced visual region information from multiple views. More specifically, a novel adaptive gating self-attention module is carefully designed to extract representations of visual regions and words, allowing the model to adaptively capture context information by regulating the internal information flow. Later, a summarization module incorporating diversity regularization is introduced serving to aggregate region-level features into image-level features from diverse perspectives. Finally, a multi-view matching scheme is elaborately devised to match multi-view image features with their corresponding text features.

Relation Reasoning. In addition to extracting salient information to reinforce semantic features, it is equally significant to build up the relationships of regions, which can excavate latent clues for complex image–text matching patterns. For example, Li et al. take the semantic relationship into account to propose a Visual Semantic Reasoning Network (VSRN [13]) that establishes the relationship across objects to capture the key objects within the given images. The architecture of VSRN is depicted in Fig. 6. Technically, VSRN starts with forging

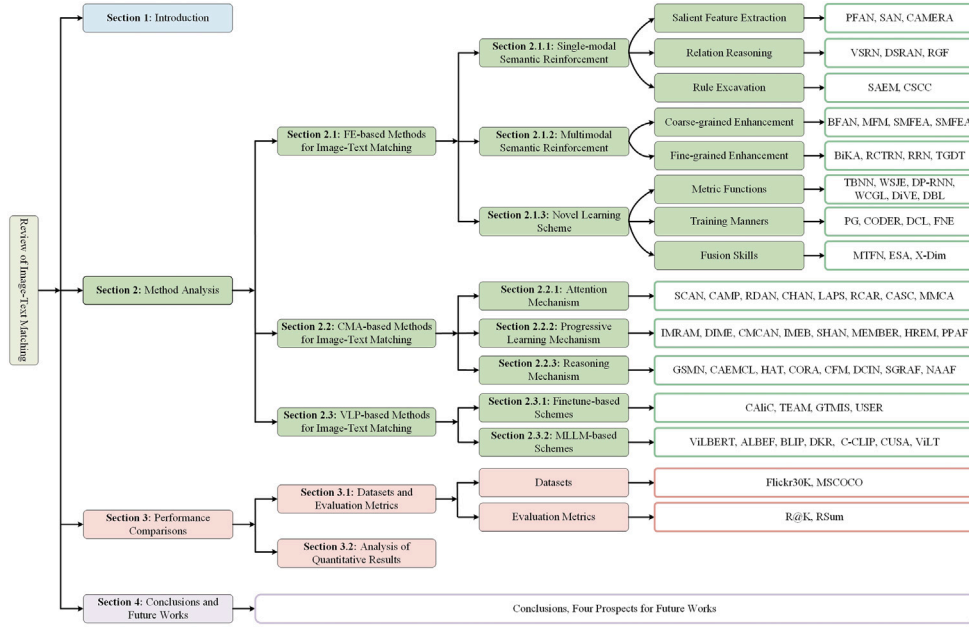


Fig. 1. Overview of the paper structure.

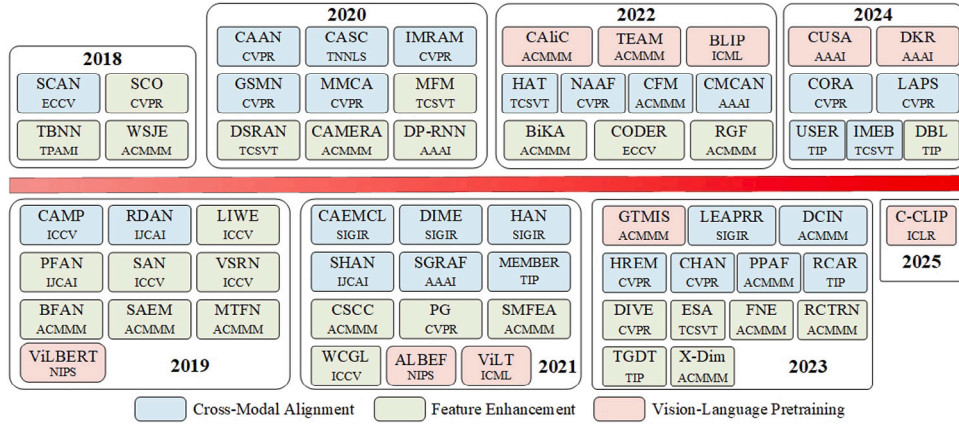


Fig. 2. Timelines of the image-text matching methods.

connections among image regions and executing reasoning processes to produce features with semantic relationships. Subsequently, VSRN brings in a gate and memory mechanism to perform global semantic reasoning on the relationship-enhanced features, which can identify the discriminative information and progressively generate the representation for the entire scene. Although VSRN can achieve promising results, it just considers coarse-grained alignment (e.g., image and text) while overlooking aligning the cross-modality data in a fine-grained manner (e.g., image regions and text words). In light of this consideration, Wen et al. put forward an innovative multi-level semantic relations enhancement approach, Dual Semantic Relations Attention Network termed DSRAN [15], to highlight the relations between regions and the relations between regional and global concepts to obtain a more accurate visual representation. For achievement, DSRAN mainly integrates two modules, including a separate semantic relations module and a joint semantic relations module. The former is predominantly responsible for capturing the semantic relations of image regions by adopting two separate graph attention networks to learn pixel-wise semantic relations and regional relations collaboratively. The latter accounts for finding the semantic relations between local objects and global pixel-wise concepts, which can be mined by using a unified graph attention network. Moreover, considering region features is short

of context information, which makes it difficult to match the words whose meanings contain global concepts. Wu et al. creatively propose an RGF [18] architecture to fuse the region features and grid features with a two-step interaction strategy so that a more comprehensive image representation can be extracted. In the first step, RGF constructs a joint graph subject to spatial information constraints, where both region features and grid features are depicted as graph nodes. Information can be transmitted along the edges by modeling the relationships utilizing the joint graph. In the second step, a cross-attention gated fusion module is put forward to delve deeper into the intricate interactions between region features and grid features, adaptively combining different types of features.

Rule Excavation. Besides, numerous works have attempted to take advantage of the interior rules of languages to assist the learning of visual features. For instance, Wu et al. propose Self-Attention Embeddings (SAEM [14]) to explore fragment relations in texts through a self-attention mechanism, aggregating fragment information into visual and textual embeddings. First of all, SAEM individually extracts salient image regions and sentence fragments using bottom-up attention and WordPiece tokens. Subsequently, the subtle and fine-grained fragment relation across image and text is built based on

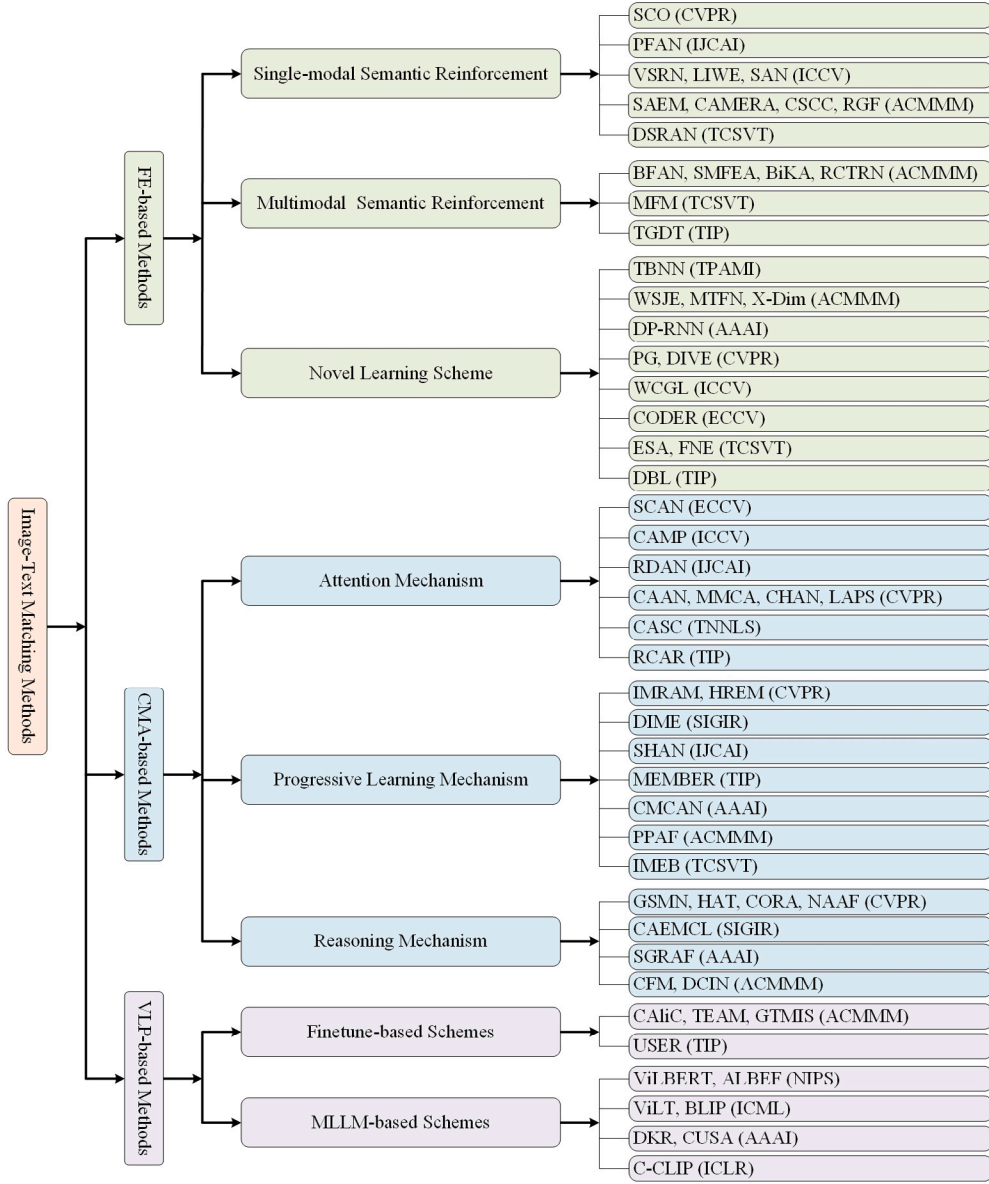


Fig. 3. Summary of methods for image-text matching task.

self-attention layers. Finally, the self-attention mechanism can discover the fragment relations and semantically salient regions separately, making the cross-modal interaction captured more accurately. Concurrently, Wehrmann et al. design a novel character-based word-embedding approach, dubbed LIWE [10] from the perspective of character level, which can project similar words across languages into the same word-embedding space. Benefiting from using character-based word-embedding for cross-modal retrieval, the storage necessities for a text encoder decrease substantially, yielding lighter and more scalable retrieval architectures. However, these methods cannot handle the challenge of unconsidered syntactical correspondence. To this aim, Zeng et al. propose a Conceptual and Syntactical Cross-modal Alignment with Cross-level Consistency (CSCC [17]) to simultaneously explore the multiple-level cross-modal alignments across the concept and syntactical with a consistency constraint, inspiring from the theory: *a sentence is not only a set of words or phrases but also a syntactic structure, consisting of a set of basic syntactic tuples*. The pivotal contributions of the CSCC can be classified into three parts. A conceptual-level cross-modal alignment is engaged for exploring the fine-grained correspondence. A syntactical-level cross-modal alignment is dedicated to explicitly learning a high-level syntactic similarity function. An empirical cross-level

consistent attention loss is designed to maintain the consistency between cross-modal attentions obtained from the above two cross-modal alignments.

2.1.2. Multimodal semantic reinforcement

In general, merely enhancing the single-modal features may trap the model in the dilemma of semantic hierarchical inconsistency. Regarding this consideration, it is necessary to reinforce the representation abilities of both image and text. To this end, numerous works are devoted to the solution of multimodal semantic reinforcement [19–24]. Current works can be roughly classed into **Coarse-grained Enhancement** and **Fine-grained Enhancement** two classes.

Coarse-grained Enhancement. The approaches in this class usually accommodate the local and global semantic embeddings. Liu et al. present a novel Bidirectional Focal Attention Network, BFAN [19], which enables the model to divert all the attention into the relevant fragments. BFAN pre-assigns attention to the inter-modality relations, identifies relevant fragments based on intra-modality relations, and jointly applies the image-to-text and text-to-image directions to avoid preference for long texts or complex images. Ma et al. take the multiple

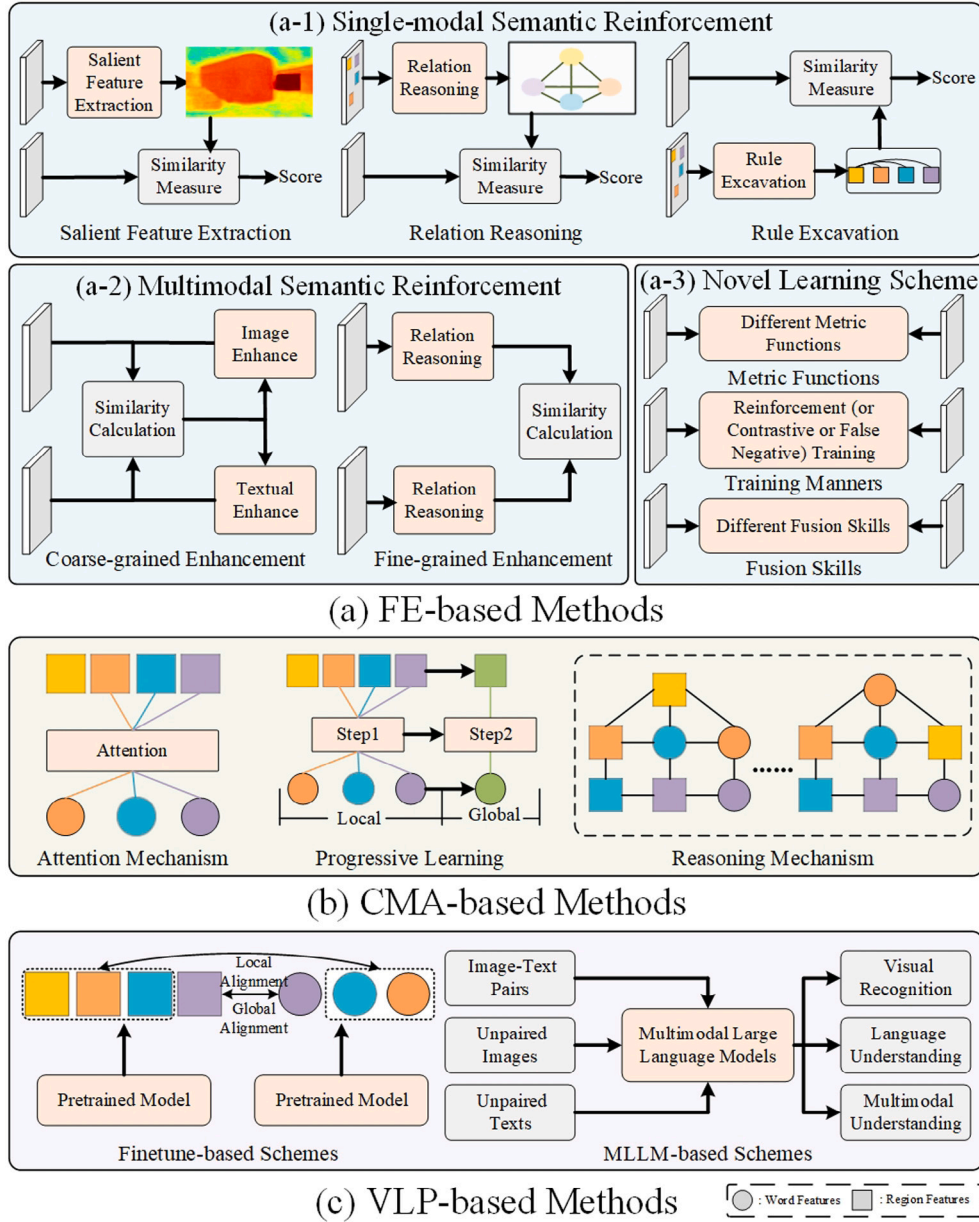


Fig. 4. Illustration of simplifying method structures.

representations of image and text modalities into account and propose an innovative multimodal matching model, termed MFM [20], which can not only make full use of the matching relationship between image and sentence based on their multi-faceted representations, but also is capable of simultaneously exploiting the relationship between multi-faceted representations. Ge et al. address the challenge of semantic misalignment based on the proposed SMFEA [21] from two aspects: constructing intrinsic structure among the fragments within respective modalities and excavating explicit structure and semantic correspondence across modalities. The core thought of SMFEA is to create a new multi-modal structured module with a shared context-aware referral tree for jointly and explicitly learning the visual-textual embedding and the cross-modal alignment. Technically, SMFEA is accomplished by two encoders, a Visual Context-aware Structured Tree encoder (VCS-Tree) and a Textual Context-aware Structured Tree encoder (TCS-Tree), which account for the modeling of the visual-textual fragment relations and jointly learning and optimizing the visual and textual features.

Fine-grained Enhancement. The approaches in this class tend to establish fine-grained dependencies between objects, which can break

through the upper-bound performance of the image-text matching. Zhu et al. elaborate a BiKA [22] (**B**idirectional **K**nowledge-assisted embedding and **A**ttention-based generation) to establish dependencies between objects using a bidirectional graph convolutional neural network, achieve cross-modal information mutual transformation by engaging a bidirectional attention-based generative network. Specifically, a knowledge graph is used for local matching to constrain the local expression of the modalities, whereas the generative network is used for mutual transformation to constrain the global expression of the modalities. However, the methods based on graph convolutional neural networks usually achieve unsatisfactory performance. With the rapid development of the Transformers, Li et al. instead plan to introduce Transformers to tackle the image-text matching task, which can pioneer a new way of thinking for this task. The challenge in introducing Transformers is a large number of parameters and an exponential computational cost. In terms of this consideration, they propose an RCTRNN [23], which can considerably achieve the trade-off between the computational budget and performance. RCTRNN employs a two-step strategy to focus on feature representation and data distribution

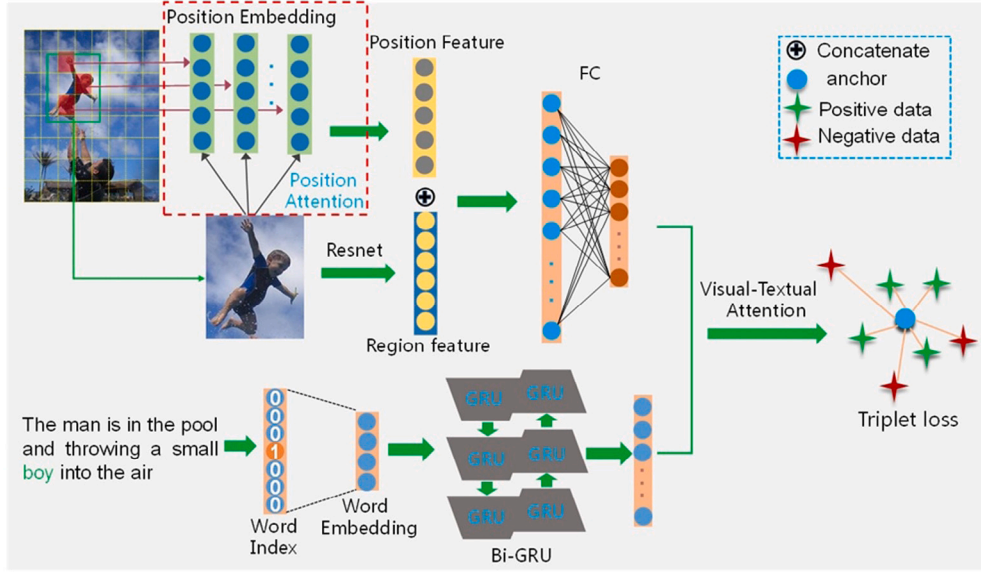


Fig. 5. A representative approach in the FE-based direction. PFAN [11] first extracts salient features for feature enhancement.

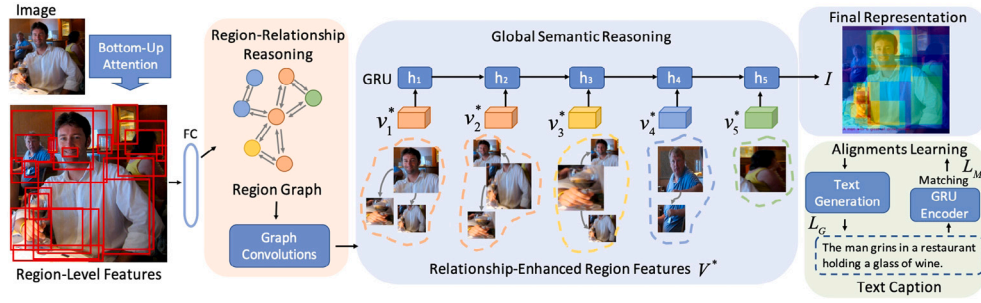


Fig. 6. A representative approach in the FE-based direction. VSRN [11] first reasons the object relationships for feature enhancement.

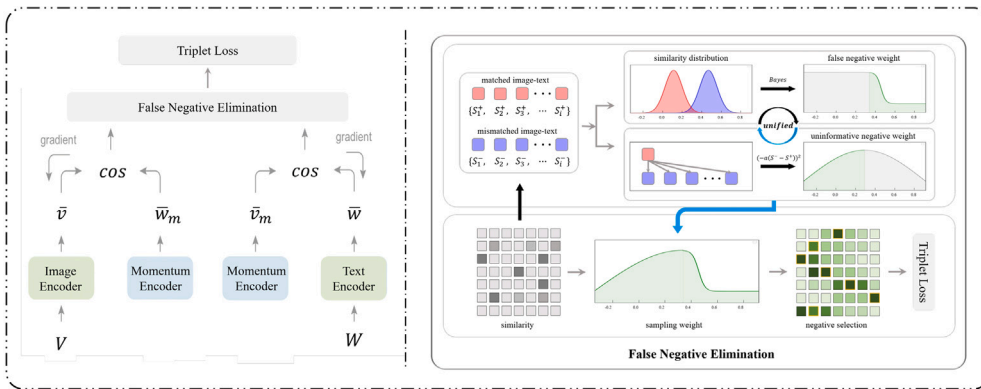


Fig. 7. A representative approach in the FE-based direction. FNE [34] explores another training manner for feature enhancement.

of different modalities, respectively. Specifically, RCTRNN begins with sending visual and textual features via a unified meshed reasoning module, effectively encoding multi-level feature relationships with prior knowledge and aggregating complementary outputs. RCTRNN ends up optimizing feature memory connections across different stages and

tackling the data distribution mismatch issue from the unified scheme by a proposed Reservoir Reasoning Network (RRN). Rather than just establishing fine-grained dependencies, it is equally important to explore the close relations between coarse- and fine-grained representations. To this end, Liu et al. exhibit a Token Guided Dual Transformer, termed

Table 1
Advantages and disadvantages of methods based feature enhancement.

Methods	Advantages	Disadvantages
(a) Single-modal semantic reinforcement		
SCO [9]	Providing visual semantics	Ignoring salient features
PFAN [11]	Learning position relations	Lacking modality interaction
SAN [12]	Learning fine-grained correlations	Hard to semantic alignment
VSRN [13]	Establishing region relations	Ignoring fine-grained alignment
SAEM [14]	Exploring fragment relations	Unconsidered syntactical correspondence
DSRAN [15]	Highlighting cross-modal relations	Lacking context information
CAMERA [16]	Summarizing multi-view regions	Overlooking region interaction
RGF [18]	Extracting comprehensive visual features	Inable to learn language rules
LIWE [10]	Using character-based word-embedding	Fail to learn syntactical correspondence
CSCC [17]	Exploring multi-level cross-modal alignments	Semantic hierarchal inconsistency
(b) Multimodal semantic reinforcement		
BFAN [19]	Divert attention into relevant fragments	Semantic misalignment
MFM [20]	Fully use matching relationship	Merely coarse-fined alignment
SMFEA [21]	Excavate explicit structure	Built upon coarse-fined alignment
BiKA [22]	Cross-modal mutual transformation	Unsatisfactory performance
RCTRN [23]	Balance between complexity and performance	Lack close relations
TGDT [24]	Combine coarse- and fine-grained representation	Limited performance
(c) Novel learning scheme		
TBNN [25]	Produce varied representations	Learned on small-scale datasets
WSJE [26]	Augment visual-semantic joint embedding	Ignore inter-objects connections
MTFN [27]	Learn a accurate similarity function	Model complex
DP-RNN [28]	Symmetrically process images and texts	Insufficient measurement metrics
PG [29]	Optimize evaluation metrics	Constrained loss function
WGL [30]	Access the global covariance	Cross-modal semantic ambiguity
CODER [31]	Optimize contrastive learning	Inadequate triplet loss
DIVE [32]	Novel similarity function	Ineffective training skill
ESA [33]	Fuse multimodal features	Independently aggregate correspondences
FNE [34]	Circumvent false negatives	Lack fusion technique
X-Dim [35]	Mine semantic dependencies	Rely limited backbones
DBL [36]	Integrate anchor-target branch	Ineffective learning strategy

Table 2
Advantages and disadvantages of methods based on cross-modal alignment.

Methods	Advantages	Disadvantages
(a) Attention mechanism		
SCAN [37]	Contextualize full latent alignments	Lack abstract concepts
CAMP [38]	Alternatively attend to region-word	Merely consider concrete objects
RDAN [39]	Adaptively find connections	Abundant computation budget
CASC [41]	Provides the global similarity	Exploit information once
MMCA [42]	Novel cross-attention	Lack information reuse
CHAN [44]	Mine meaningful alignments	Visual redundancy and ambiguity
RCAR [45]	Contextualize and aggregate representations	Over computation complexity
LAPS [46]	Discern and rectify visual patches	parameter ineffectiveness
(b) Progressive learning mechanism		
IMRAM [47]	Capture sophisticated patterns	Rely on expert experience
DIME [48]	Achieve unified and dynamic interactions	Just focus on region-word pairs
SHAN [49]	Multi-step cross-modal reasoning	Ignore instance-level interactions
MEMBER [50]	Fine-grained alignment and fusion	Lack other interactions
CMCAN [51]	Infer the matching confidence	Inbalance between interaction and efficacy
HREM [52]	Capture multi-level relations	Limited local associations
PPAF [53]	Model association uncertainty	Without relation reasoning
IMEB [54]	Balance between interaction and efficiency	Neglect hierarchical reasoning
(c) Reasoning mechanism		
GSMN [55]	Fine-grained correspondence	Cannot express object information
CAEMCL [56]	Shared semantic concepts	Limited performance representation
SGRAF [57]	Model similarity reasoning	Ignorance of mismatched fragments
HAT [58]	Facilitate semantic integrity	Lack causal inference
NAAF [59]	Consider mismatched fragments	Subject to inadequate backbones
CFM [60]	Employ causal inference	Underexplore matching score
DCIN [61]	Learn causality	Underexplore the importance of matching score
CORA [62]	Efficiently encode multiple semantic relations	Lack similarity reasoning

Table 3
Advantages and disadvantages of methods based on vision-language pretraining.

Methods	Advantages	Disadvantages
(a) Finetune-based schemes		
CALiC [64]	Improve unimodal encoder	Contain abstract information
TEAM [65]	Explicitly supervise alignments	Assume abstract information
GTMIS [63]	Adopt weak alignment	Inter-, intra-modal matching loss
USER [68]	Differentially treat regions	Constrained model accuracy
(b) MLLM-based schemes		
ViLBERT [69]	Learning task-agnostic joint representations	Limited retrieval performance
ALBEF [70]	Densely align image text modalities	Performance-efficiency imbalance
ViLT [71]	Simplified feature extraction	Unsatisfactory supervision mode
BLIP [72]	Flexible transfer ability	Causing catastrophic forgetting
DKR [73]	Dynamically filter incorrect old knowledge	Limited retrieval performance
CUSA [74]	Solve matching and semantic loss	Training difficult
C-CLIP [75]	Multimodal continual learning	Inadequate task verification

TGDT [24], from a novel perspective by combining coarse- and fine-grained representation learning into a unified framework. TGDT is individually integrated by two homogeneous branches for image and text modalities, which are responsible for incorporating and leveraging both coarse- and fine-grained retrievals into a unified framework.

2.1.3. Novel learning scheme

In addition to reinforcing the visual and textual features, additional works are aiming at designing new schemes to further improve the model performance, e.g., devising appropriate **metric functions** [25, 26, 28, 30, 32, 36], providing more effective **training manner** [29, 31, 34], and robust **fusion skills** [27, 33, 35].

Metric Functions. The expectation of devising appropriate metric functions is to measure the features captured from the image and text effectively. In light of this consideration, Wang et al. take the lead in proposing a TBNM [25], which consists of an embedding network and a similarity network, accounting for the learning of an explicit shared latent embedding space with a maximum-margin ranking loss and the fusion of the outputs from the two branches with a regression loss to predict a similarity score. Mithun et al. propose WSJE [26] to unified exploit web images and corresponding tags along with fully annotated datasets to learn the visual-semantic joint embedding and augment a typical supervised pair-wise ranking loss with weakly-annotated web images. Benefiting from this, WSJE can resolve the challenge that current methods are trained with small-scale datasets covering a limited number of images with ground-truth sentences. However, these methods fail to build up the connection between the objects that are semantically related. Chen et al. propose a Dual Path Recurrent Neural Network (DP-RNN [28]) to process images and sentences symmetrically. DP-RNN is in an attempt to reorder the image objects based on the positions of their most related words in the text. Meanwhile, the same way as extracting the hidden features from word embeddings is applied to extract high-level object features from the reordered object inputs. Although the approaches above have obtained remarkable performance, the measurement taking place on the embedding features cannot well characterize the intrinsic structures to difficult to achieve robust and crucial similarity measurement. As a remedy, Wang et al. propose a Wasserstein Coupled Graph Learning (WCGL [30]) to make the model access the global covariance of all samples within one or across different ones with the help of graphs. To be specific, WCGL starts with constructing graphs separately based on two modalities and then passing the constructed graphs through correlated graph encoders for the extraction of robust features. Subsequently, a Wasserstein coupled dictionary is constructed for further feature learning, where the Wasserstein coupled dictionary contains multiple pairs of counterpart graph keys with each key corresponding to one modality. Lastly, the input graphs are further transformed into the dictionary space to facilitate the similarity measurement by using the constructed dictionary. In terms of the inherent ambiguity of cross-modal semantic information, Kim et al. present a novel set-based embedding method

namely DiVE [32] distinct from previous work in two aspects. One is a new similarity function called smooth-Chamfer similarity to alleviate the side effects of existing similarity functions for set-based embedding. The other is a novel set prediction module to produce a set of embedding vectors that effectively captures the diverse semantics of the input. Moreover, Diao et al. aim to boost intrinsic matching patterns by utilizing the knowledge transfer between peer branches and present a brand-new Deep Boosting Learning (DBL [36]) algorithm integrating an anchor branch and a target branch, which can provide insights into the data properties and gain more advanced knowledge to develop optimal features and distance metrics, respectively. The anchor branch learns the absolute or relative distance between positive and negative pairs, which can provide a foundational understanding of the particular network and data distribution. Concurrently, the target branch is tasked with more adaptive margin constraints for the further enlargement of the relative distance between matched and unmatched samples.

Training Manners. The manner of effective training aims to explore another training scheme to replace the traditional solution that implicitly transfers the modalities into a unified space for distance measurement. For this purpose, Yan et al. advocate designing a novel attention-supervised mechanism, namely PG [29], to directly optimize the evaluation metrics within the unified space. Although PG is a kind of supervised attention, it requires no additional annotations benefited by a novel Discrete-continuous action space policy gradient algorithm, which is capable of effectively modeling complex action spaces. As with PG, Wang et al. also aim to explore a novel training scheme for further improving the potential of cross-modal matching and attempt to introduce prevailing contrastive learning mechanisms. However, trivially introducing the contrastive learning mechanisms is sub-optimal as the manual weighting of negative pairs and unawareness of external knowledge. As a remedy, a novel coupled diversity-sensitive momentum contrastive learning (CODER [31]) is proposed to improve cross-modal representation. To achieve the proposed CODER, a novel Diversity-Sensitive Contrastive Learning (DCL) with two branches is elaborately designed. The first branch learns instance-level embeddings from image/text and generates pseudo-online clustering labels. The second branch is queried from the commonsense knowledge graph for concept-level descriptors. Besides, dynamic dictionaries are used for both modalities to expand image-text pairs, and achieve diversity-sensitiveness through adaptive negative pair weighting. All of the above methods adopt triplet loss as the optimization objective, which easily traps the model in the dilemma of employing the most similar yet true samples as hard negatives. To this aim, Li et al. propose a novel False Negative Elimination (FNE [34]) strategy to select negatives via sampling, which could alleviate the problem introduced by false negatives. The architecture of FNE is depicted in Fig. 7. Specifically, positive and negative sample distributions are first established by separately using their similarities to the anchor. Then, the false-negative probability of a sample according to its similarity to the anchor and the above distributions is calculated based on the rule of Bayes, where

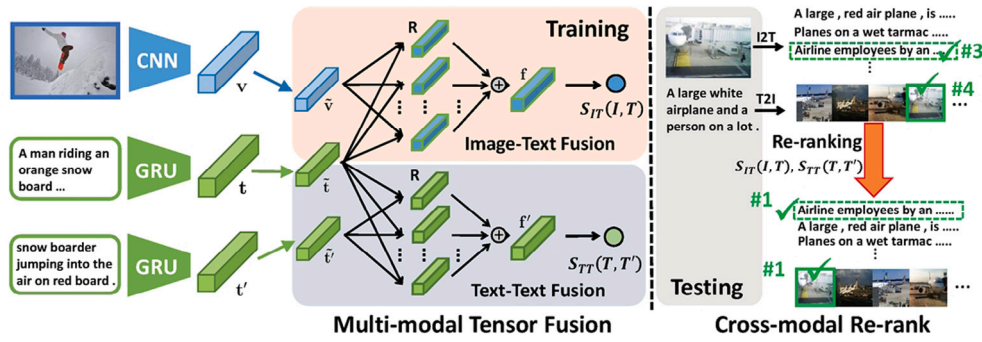


Fig. 8. A representative approach in the FE-based direction. MTFN [27] explores rank-based tensor fusion for feature enhancement.

the probability is used as the sampling weight in negative sampling. Finally, a momentum-based memory module is designed to hold a large negative buffer and apply our negative sampling strategy across it. Moreover, to make the model focus on hard negatives, a cut-down strategy is used to reassign sampling weights for simple negatives.

Fusion Skills. The reason for designing robust fusion skills is also viewed as a novel learning scheme because it changes the learning direction to eliminate the cross-modal gaps to some extent. For example, Wang et al. propose a novel Multi-modal Tensor Fusion Network (MTFN [27]) to explicitly learn an accurate image-text similarity function with rank-based tensor fusion, replacing seeking a common embedding space for each image-text instance in the training phase. For the testing phase, they deploy a general cross-modal re-ranking scheme to refine the matching results. The architecture of MTFN is depicted in Fig. 8. Rather than fusing tensors, Zhu et al. design a plug-and-play External Space attention Aggregation (ESA [33]) module to make the fusion of modal features under spatial dimensional attention, which enables the trade-off to balance effectiveness and efficiency. The core of ESA is to integrate discriminative information and the aggregation of local features into a unified space for fine-grained representation. Although satisfactory results are obtained step-by-step, independently aggregating all dimensional correspondences of cross-modal features like the above approaches while ignoring intrinsic dependencies among dimensions probably leads to inferior aggregation for semantic similarity as the dimensions are not mutually independent. To solve this issue, Zhang et al. propose a novel cross-dimensional semantic dependency-aware model, X-Dim [35], to explicitly and adaptively mine the semantic dependencies between dimensions in a shared space. Concretely, X-Dim designs a generalized framework to learn semantic dependency degrees of dimensions and develop adaptive sparse probabilistic learning to make the model capture precise dependencies autonomously.

2.2. CMA-based methods for image-text matching

The solutions lying in the cross-modal alignment expect to interact visual features with corresponding textual features to maximize the similarity degrees of given image-text pairs. Existing CMA-based methods can be also roughly categorized into three classes based on the differences of learning mechanisms, i.e., *attention mechanism* [37–46], *progressive learning mechanism* [47–54], and *reasoning mechanism* [55–62].

2.2.1. Attention mechanism

Considering the promising discrimination ability for the salient information from redundant information, the attention mechanism has become a prevalent tool for various tasks, ranging from computer vision and natural language processing.

Traditional Attention. Prior works intend to introduce traditional cross-attention mechanisms to make the textual word focus on corresponding visual regions or vice versa. For instance, different from

traditional methods that simply aggregate the similarity of all possible region-word pairs without attending differentially to more and less significant words or regions, Lee et al. develop the traditional cross-attention and present a novel Stacked Cross Attention Network (SCAN [37]) to infer image-text similarity by contextualizing the full latent alignments using both image regions and sentence words. SCAN consists of an image-to-text formulation and a text-to-image formulation, where the bidirectional formulation is inherited by following attention mechanism-based methods. For the image-to-text formulation, the sentence words attend with respect to each region, compare each image region to the attended information from the sentence, and decide the importance of the image regions. Similarly, the image regions attending to each word and deciding whether to pay more or less attention to each word are performed on the text-to-image formulation. Similar to SCAN, Wang et al. propose Cross-modal Adaptive Message Passing (CAMP [38]) to adaptively control the information flow for message passing across modalities, where the foundation of CAMP is also the cross-attention mechanism. The differences between SCAN and CAMP could come down to the motivation and implementation details. The motivation for CAMP comes from human cognition that alternatively attends to regions in images and words in sentences, selecting the most salient information considering the interaction between both modalities, while SCAN aims at aggregating all possible region-word pairs. As for the implementation details, besides taking comprehensive and fine-grained cross-modal interactions into account, CAMP properly handles negative pairs and irrelevant information with an adaptive gating scheme and proposes a hardest negative binary cross-entropy loss for training based on fused features. Although impressive performances, the above methods only consider the concrete objects, failing to mine abstract concepts (e.g., some actions). As a remedy, Hu et al. propose a Relation-wise Dual Attention Network (RDAN [39]) to alternatively find the implicit and intrinsic connections between concrete objects and abstract concepts. Particularly, RDAN begins with encoding local correlations and the global dependencies between regions and words built upon an over-complete set containing image-text pairs. Then, RDAN elaborates on a dual pathway attention network to infer the visual-semantic alignments and image-text similarity.

Effective Attention. Although remarkable performances are obtained with the assistance of the cross-attention mechanism, the dot-product calculation behind the cross-attention mechanism may bring redundant or irrelevant region-word alignments, deteriorating retrieval accuracy, and limiting efficiency. With this in mind, substantial works shift their attention to the exploration of effective attention mechanisms [44–46]. For example, Pan et al. attempt to learn fine-grained image-text matching from the perspective of information coding and propose Cross-modal Hard Aligning Network (CHAN [44]) to mine meaningful alignments. To be explicit, CHAN not only discards redundant alignments and better discovers the granular semantic correspondence, but also relieves the costly dense cross-attention matching, thus significantly improving cross-attention baselines both in accuracy and efficiency. Apart from CHAN, Fu et al. plan to supplement patch-word

Table 4
Performances of FE-based Methods on MSCOCO1K dataset.

	Methods	Publications	Text retrieval			Image retrieval			RSum
			R@1	R@5	R@10	R@1	R@5	R@10	
(a) Single-modal semantic reinforcement									
1	SCO [9]	CVPR2018	66.6	91.8	96.6	55.5	86.6	93.8	490.9
2	PFAN [11]	IJCAI2019	76.5	96.3	99.0	61.6	89.6	95.2	518.2
3	SAN [12]	ICCV2019	85.4	97.5	99.0	69.1	93.4	97.2	541.6
4	VSRN [13]	ICCV2019	76.2	94.8	98.2	62.8	89.7	95.1	516.8
5	SAEM [14]	ACMMM2019	71.2	94.1	97.7	57.8	88.6	94.9	504.3
6	DSRAN [15]	TCSVT2020	80.6	96.7	98.7	64.5	90.8	95.8	527.1
7	CAMERA [16]	ACMMM2020	77.5	96.3	98.8	63.4	90.9	95.8	522.7
8	RGF [18]	ACMMM2022	83.7	97.7	99.1	68.4	92.8	97.5	539.2
9	LIWE [10]	ICCV2019	73.2	95.5	98.2	57.9	88.3	94.5	507.6
10	CSCC [17]	ACMMM2021	78.8	96.1	99.0	66.6	92.5	96.4	529.4
(b) Multimodal semantic reinforcement									
11	BFAN [19]	ACMMM2019	74.9	95.2	–	59.4	88.4	–	–
12	MFN [20]	TCSVT2020	58.9	86.3	92.4	47.7	81.0	90.9	457.2
13	SMFEA [21]	ACMMM2021	75.1	95.4	98.3	62.5	90.1	96.2	517.6
14	BiKA [22]	ACMMM2022	77.6	96.5	98.6	62.8	90.3	95.8	521.6
15	RCTRN [23]	ACMMM2023	79.4	96.6	98.3	66.9	92.2	96.8	530.2
16	TGDT [24]	TIP2023	79.6	96.9	99.0	66.7	92.2	97.0	531.4
(c) Novel learning scheme									
17	TBNN [25]	TPAMI2018	54.0	84.0	91.2	43.3	76.8	87.6	436.9
18	WSJE [26]	ACMMM2018	61.5	–	96.1	46.3	–	89.4	–
19	MTFN [27]	ACMMM2019	74.3	94.9	97.9	60.1	89.1	95.0	511.3
20	DP-RNN [28]	AAAI2020	75.3	95.8	98.6	62.5	89.7	95.1	517.0
21	PG [29]	CVPR2021	84.0	95.8	97.8	63.9	88.9	95.6	526.0
22	WCGL [30]	ICCV2021	75.4	95.5	98.6	60.8	89.3	95.3	514.9
23	CODER [31]	ECCV2022	82.1	96.6	98.8	65.5	91.5	96.2	530.7
24	DIVE [32]	CVPR2023	80.6	96.3	98.8	64.7	91.4	96.2	528.0
25	ESA [33]	TCSVT2023	80.4	96.5	98.8	64.2	91.3	96.3	527.5
26	FNE [34]	TCSVT2023	82.5	96.6	99.0	67.7	90.5	95.2	531.5
27	X-Dim [35]	ACMMM2023	80.9	96.9	98.9	64.7	90.9	96.5	528.8
28	DBL [36]	TIP2024	88.8	99.0	–	76.1	95.5	–	–

Table 5
Performances of FE-based methods on MSCOCO5K dataset.

Methods		Publications	Text retrieval			Image retrieval			RSum
			R@1	R@5	R@10	R@1	R@5	R@10	
(a) Single-modal semantic reinforcement									
1	SCO [9]	CVPR2018	42.8	72.3	83.0	33.1	62.9	75.5	369.6
2	SAN [12]	ICCV2019	65.4	89.4	94.8	46.2	77.4	86.6	459.8
3	VSRN [13]	ICCV2019	53.0	81.1	89.4	40.5	70.6	81.1	415.7
4	DSRAN [15]	TCSVT2020	57.9	85.3	92.0	41.7	72.7	82.8	432.4
5	CAMERA [16]	ACMMM2020	55.1	82.9	91.2	40.5	71.7	82.5	423.9
6	RGF [18]	ACMMM2022	63.5	87.9	93.5	46.8	76.1	85.1	452.9
7	CSCC [17]	ACMMM2021	55.6	83.6	91.2	40.8	73.2	84.3	428.7
(b) Multimodal semantic reinforcement									
8	SMFEA [21]	ACMMM2021	54.2	–	89.9	41.9	–	83.7	–
9	BiKA [22]	ACMMM2022	54.5	83.4	91.4	40.2	70.9	80.7	421.1
10	RCTRN [23]	ACMMM2023	57.1	83.4	91.9	43.6	71.9	83.7	431.6
11	TGDT [24]	TIP2023	59.3	85.7	92.8	45.1	75.2	85.1	443.2
(c) Novel learning scheme									
12	MTFN [27]	ACMMM2019	48.3	77.6	87.3	35.9	66.1	76.1	391.3
13	PG [29]	CVPR2021	68.7	88.7	93.0	46.2	77.8	85.5	459.9
14	DIVE [32]	CVPR2023	60.4	86.2	92.4	42.6	73.1	83.1	437.8
15	ESA [33]	TCSVT2023	59.1	85.5	92.1	41.8	72.3	82.8	433.6
16	FNE [34]	TCSVT2023	64.3	87.0	92.9	47.6	74.9	83.5	450.2
17	X-Dim [35]	ACMMM2023	59.3	86.0	92.5	43.2	71.9	82.4	435.3
18	DBL [36]	TIP2024	70.9	91.1	–	54.8	81.0	–	–

attention to address the resultant issue of visual patch redundancy and patch ambiguity for semantic alignment. To this end, they propose a novel Linguistic-Aware Patch Slimming (LAPS [46]) framework, which enables explicit identification of redundant visual patches with language supervision and rectification of their semantic and spatial information to facilitate more effective and consistent patch-word alignment. Different from the CHAN and LAPS focusing on the information

excavation, there is also other work striving to explore the parameter effectiveness existing in the cross-attention operations. Diao et al. develop an RCAR [45] to efficiently encode the message output, yielding automatically contextualized and aggregated cross-modal representations. The architecture of RCAR is depicted in Fig. 9. The achievement of RCAR is dependent on two simple but relatively effective attention regulators, namely a Recurrent Correspondence Regulator (RCR) and a

Table 6
Performances of FE-based Methods on Flickr30K dataset.

	Methods	Publications	Text retrieval			Image retrieval			RSum
			R@1	R@5	R@10	R@1	R@5	R@10	
(a) Single-modal semantic reinforcement									
1	SCO [9]	CVPR2018	44.2	74.1	83.6	32.8	64.3	74.9	373.9
2	PFAN [11]	IJCAI2019	70.0	91.8	95.0	50.4	78.7	86.1	472.0
3	SAN [12]	ICCV2019	75.5	92.6	96.2	60.1	84.7	90.6	499.7
4	VSRN [13]	ICCV2019	71.3	90.6	96.0	54.7	81.8	88.2	482.6
5	SAEM [14]	ACMMM2019	69.1	91.0	95.1	52.4	81.1	88.1	476.8
6	DSRN [15]	TCSVT2020	80.5	95.5	97.9	59.2	86.0	91.9	511.0
7	CAMERA [16]	ACMMM2020	78.0	95.1	97.9	60.3	85.9	91.7	508.9
8	LIWE [10]	ICCV2019	69.6	90.3	95.6	51.2	80.4	87.2	474.3
9	CSCC [17]	ACMMM2021	72.7	93.4	96.5	61.2	86.7	91.5	502.0
10	RGF [18]	ACMMM2022	84.5	97.3	99.0	66.8	89.7	94.3	531.6
(b) Multimodal semantic reinforcement									
11	BFAN [19]	ACMMM2019	68.1	91.4	–	50.8	78.4	–	–
12	MFM [20]	TCSVT2020	50.2	78.1	86.7	38.2	70.1	80.2	403.5
13	SMFEA [21]	ACMMM2021	73.7	92.5	96.1	54.7	82.1	88.4	487.5
14	BiKA [22]	ACMMM2022	75.2	91.6	97.4	54.8	82.5	88.6	490.1
15	RCTRN [23]	ACMMM2023	78.4	95.4	96.8	60.4	84.9	93.7	509.6
16	TGDT [24]	TIP2023	79.6	96.9	99.0	66.7	92.2	97.0	531.4
(c) Novel learning scheme									
17	TBNN [25]	TPAMI2018	43.8	72.1	80.4	33.5	62.4	73.9	366.1
18	WSJE [26]	ACMMM2018	47.4	–	85.9	35.2	–	74.8	–
19	MTFN [27]	ACMMM2019	65.3	88.3	93.3	52.0	80.1	86.1	465.1
20	DP-RNN [28]	AAAI2020	70.2	91.6	95.8	55.5	81.3	88.2	482.6
21	PG [29]	CVPR2021	82.8	95.9	97.9	62.2	89.3	93.8	521.9
22	WCGL [30]	ICCV2021	74.8	93.3	96.8	54.8	80.6	87.5	487.8
23	CODER [31]	ECCV2022	83.2	96.5	98.0	63.1	87.1	93.0	520.9
24	DIVE [32]	CVPR2023	80.9	94.7	97.6	59.4	85.6	91.1	509.3
25	ESA [33]	TCSVT2023	83.1	96.3	98.7	62.4	87.2	92.5	520.2
26	FNE [34]	TCSVT2023	85.4	98.1	99.2	70.1	90.7	94.7	538.2
27	X-Dim [35]	ACMMM2023	83.1	96.3	98.4	61.7	86.1	91.4	517.0
28	DBL [36]	TIP2024	79.1	96.4	–	60.3	84.6	–	–

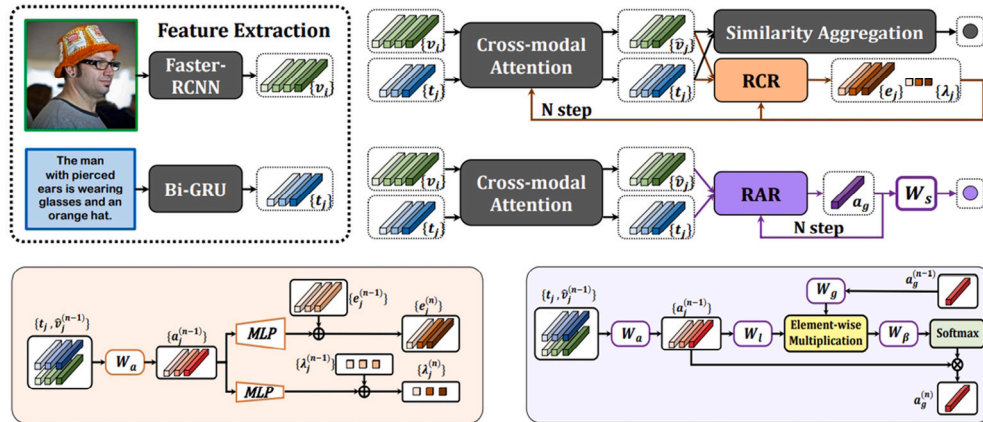


Fig. 9. A representative approach in the CMA-based direction. RCR [45] designs two effective attention structures for cross-modal alignment.

Recurrent Aggregation Regulator (RAR). RCR is in charge of facilitating the cross-modal attention unit progressively with adaptive attention factors to capture more flexible correspondence, while the RAR can adjust the aggregation weights repeatedly to increasingly emphasize important alignments and dilute unimportant ones.

Comprehensive Attention. Furthermore, numerous works aim to design more comprehensive cross-attention mechanisms for the thorough extraction of latent information. Among them, encountering the challenge of local misalignment caused by inaccurately detecting image regions or random correspondence between high-level semantics and single-image regions, Xu et al. advance a novel approach named Cross-modal Attention with Semantic Consistency (CASC [41]) to conduct cross-modal attention for local alignment and multilabel prediction for global semantic consistency. By doing so, CASC can directly extract semantic labels from the available sentence corpus without additional

labor cost, which further provides a global similarity constraint for the aggregated region-word similarity obtained by the local alignment. Wei et al. propose a novel Multi-Modality Cross Attention (MMCA [42]) network to not only model intra-modality correlation like most previous works but also establish inter-modality correspondence of image regions and sentence words in a unified deep model. MMCA integrates a novel cross-attention mechanism able to complement and enhance image regions and sentence words from each other for image and sentence matching.

2.2.2. Progressive learning mechanism

Repeated Learning. In general, it is hard to make full use of extracted information in a one-time manner, thus, some works naturally advocate repeated usage of extracted information. Based on

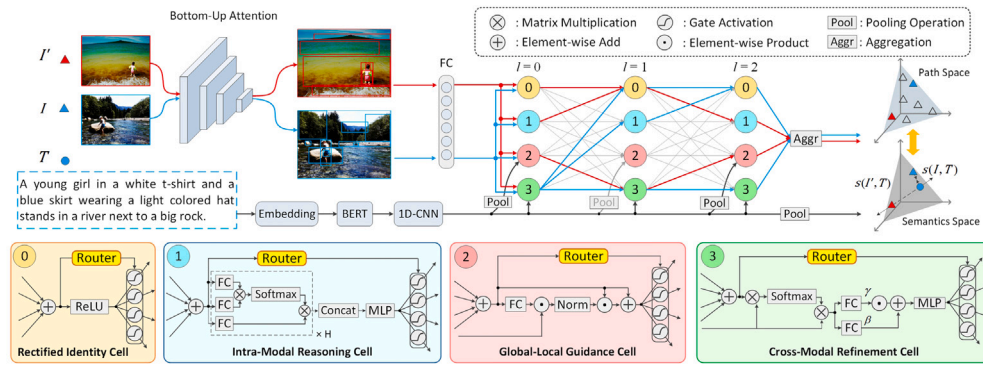


Fig. 10. A representative approach in the CMA-based direction. DIME [48] designs four types of cells for cross-modal alignment.

this, some of the current works focus on designing progressive learning mechanisms to gradually improve retrieval accuracy. Chen et al. propose an Iterative Matching with Recurrent Attention Memory (IMRAM [47]) method, in which image-text correspondences are captured with multiple steps of alignments to capture sophisticated matching patterns. Concretely, the contributions of IMRAM can be ascribed to two techniques: an iterative matching scheme to explore fine-grained correspondence progressively and a memory distillation unit to refine alignment knowledge from early steps to later ones. Current works lack flexibility as they heavily rely on expert experience and empirical feedback toward the design of interaction patterns. Qu et al. develop a novel modality interaction modeling network, DIME [48], based upon the routing mechanism, which can achieve unified and dynamic multi-modal interaction. The architecture of DIME is depicted in Fig. 10. DIME comprises four types of cells called basic units, enables the exploration of different levels of modality interactions, and enables the interactions to connect in a dense strategy to construct a routing space. Although the methods above produce remarkable results, the matching procedure only performs on the extracted region-word local pairs, which may be inconsistent/unreliable from the global perspective of image-text, resulting in inaccurate relevance measurement. To this end, Zhang et al. invent a Cross-Modal Confidence-Aware Network (CMCAN [51]) to infer the matching confidence representing the reliability of matched region-word pairs. In detail, the matching confidence is calculated through the relevance between the semantics of image regions and the completely described semantics in the image with the bridge of the given texts. Further, the region to its visual context in the image is extended for rich expression of the region semantics. Finally, local semantic similarities are weighted with the inferred confidence to filter out unreliable matched pairs for aggregation. For the achievement of a better trade-off between fine-grained interaction and retrieval efficacy, Li et al. develop a novel Independent Memory-Enhanced embedding learning framework (IMEB [54]) with a lightweight memory module to simultaneously exploit the complementarity of fine-grained interaction and retrieval efficacy. First, a cross-modal association graph is utilized to learn cross-modal fine-grained interaction information. Second, a memory-assisted embedding learning network is delicately devised to store these prototypical features after interaction as agents. Third, these agent-level prototypical features interact directly with each other so as to obtain cross-modal memory-enhanced embeddings. Benefiting from this, IMEB cooperatively cross-modal interaction information and maintains retrieval efficiency.

Hierarchical Learning. In addition to reusing cross-modal information by the heuristic learning mode, there are some other methods aiming to repeatedly learn cross-modal information hierarchically. Ji et al. pay attention to exploiting multi-level information to locate the hierarchical fine-grained relevance and present a Step-wise Hierarchical Alignment Network (SHAN [49]) to abstract a single-step image-text matching task to a multi-step cross-modal reasoning

process. Concretely, the local-to-local, global-to-local, and global-to-global alignment are sequentially achieved. As a consequence, this progressive alignment strategy supplies SHAN with more complementary and sufficient semantic clues to grasp the hierarchical image-text correlations. With a similar motivation to the SHAN, Li et al. propose a MEMBER [50] by designing Memory-based EMbedding Enhancement for image-text Retrieval for the accomplishment of fine-grained alignment and fusion. MEMBER is composed of memory banks, which are responsible for enriching image features with relevant text features stored in the text memory bank so that both mutual embedding enhancement across modalities and retrieval efficiency can be accomplished at the same time. Drawing inspiration from the common sense that high-quality cross-modal semantic alignment not only relies on fragment-level relations within a sample but also needs to pay attention to capturing instance-level interactions among samples and modalities. Therefore, Fu et al. propose an innovative Hierarchical Relation Modeling framework (HREM [52]) to explicitly capture fragment- and instance-level relations to learn discriminative and robust cross-modal embeddings. The architecture of HREM is depicted in Fig. 11. First, a novel cross-embedding association graph is proposed for explicit identification of the connection relation, on which the relevance relation between batch samples with fragment-level semantic matching can be learned. Next, two relation interaction mechanisms are advanced to explore inter-modal and intra-modal relations synchronously or asynchronously. Typically, the challenge of the huge modal gap is still unsolved perfectly because of the calculated similarity based on local associations, which underestimates the uncertainty of associations and discriminative positive associations, leading to cross-modal correspondence ambiguity. With this in mind, a novel Progressive Positive Association Framework (PPAF [53]) is presented by Li et al. who model association uncertainty as a normal distribution and progressively uncovers direct and potential positive associations based on the features of this distribution. Besides, they design a positive association matching to adaptively combine multi-step associations for local matching.

2.2.3. Reasoning mechanism

It is beneficial to make information communication between extracted local regions for perfecting semantic consistency. Therefore, previous works are in an attempt to introduce the concept of graphs to reason the relationships within or across modalities. To address the obstacle of how to learn the correspondence between image and text, Liu et al. put forward a novel Graph Structured Matching Network (GSMN [55]) to explicitly model object, relation, and attribute as a structured phrase for fine-grained correspondence. GSMN is accomplished through node-level and structure-level matching. In node-level matching, each node (which can be an object, relation, or attribute) is linked to relevant nodes from another modality. Subsequently, at the structure level, the associated nodes integrate neighborhood associations to jointly deduce fine-grained correspondences. As with GSMN, He et al. present a CAEMCL [56] with cross-graph attention to explicitly

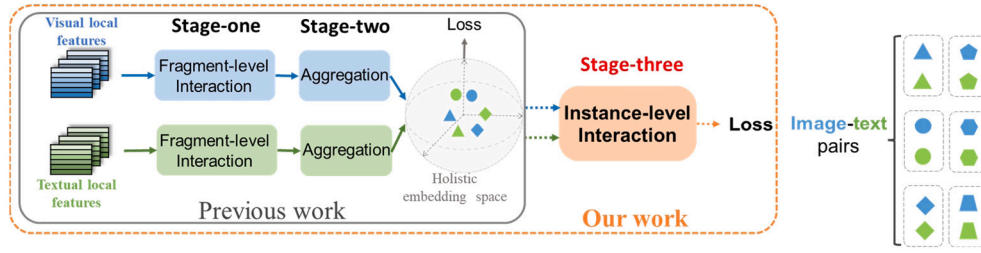


Fig. 11. A representative approach in the CMA-based direction. HREM [52] effectively achieve hierarchical feature learning for cross-modal alignment.

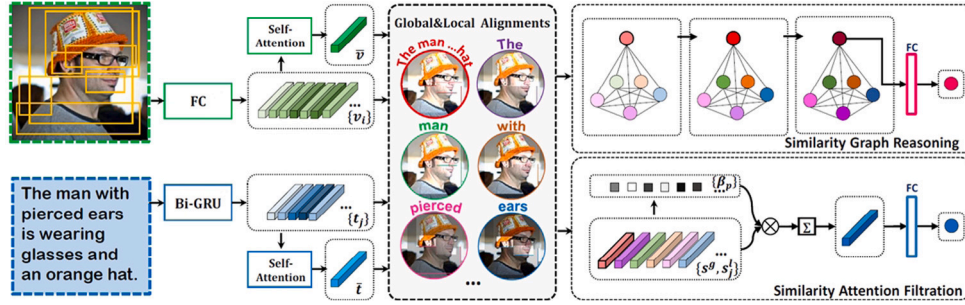


Fig. 12. A representative approach in the CMA-based direction. SGRAF [57] introduces graph reasoning on the calculated similarity scores for cross-modal alignment.

model shared semantic concepts, which can be well utilized to guide the feature learning process of each modality and promote the common embedding learning. Due to the ignorance that semantics cannot be accurately expressed by object information alone, Dong et al. propose a HAT [58] being the foundation of graph convolutional networks to hierarchically gather information to facilitate the integrity of object semantics. Recently, Pham et al. introduce scene graphs to represent captions with nodes for objects and attributes interconnected by relational edges for computational lightweight. They present a Composition model for Object Relations and Attributes, termed CORA [62], to efficiently encode object-attribute and object-wise semantic relations, offer the ability to utilize the strong relational inductive bias of graph neural networks to learn object-attribute and object-wise relations effectively, and result in a robust and fast-performing system.

The function of using graph reasoning is not merely to guarantee semantic integrity, there also exist other benefits after adopting it, such as causal inference [60,61], similarity enhancement [57,59], and so on. Recent studies employ hard negative mining to capture the multiple correspondences between visual-textual domains. However, truly informative negative samples are scarce in training data and difficult to acquire from randomly sampled mini-batches. Inspired by causal inference, Wei et al. propose a Counter Factual Matching (CFM [60]) mainly containing three components, *i.e.*, a gradient-guided feature selection, a self-exploration, and a self-adjustment with respect to the automatic causal factor identification, causal factor completeness, and counterfactual sample synthesis. At the same time, Li et al. invent an innovative Deconfounded Causal Inference Network (DCIN [61]) to repair the damages from intra- and inter-modal confounders with two major functions. First, DCIN decomposes intra- and inter-modal confounders and integrates them into the encoding of visual and textual features, effectively removing spurious correlations in image-text matching. Second, it employs causal inference to reduce biases from external knowledge. As a result, DCIN can learn causality rather than spurious correlations stemming from dataset bias. Encountering the challenge of how to make the most of these alignments to infer more accurate matching scores is still underexplored, Diao et al. take the lead in introducing graph reasoning on the calculated similarity scores and propose a Similarity Graph Reasoning and Attention Filtration (SGRAF [57]). The architecture of SGRAF is depicted in

Fig. 12. SGRAF is built upon learned vector-based similarity representations and integrated by two modules, named Similarity Graph Reasoning (SGR) and Similarity Attention Filtration. The two modules are in charge of inferring relation-aware similarities and selectively integrating significant alignments, respectively. Sequentially, Zhang et al. further exploit the positive effect of matched fragments and the negative effect of mismatched fragments to reason a more rational matching score. They propose a Negative-Aware Attention Framework (NAAF [59]) with two main contributions. First of all, it elaborately designs an iterative optimization approach to fully extract mismatched fragments, enhancing discriminative and robust negative effects. Secondly, it creates a two-branch matching mechanism, which accurately computes similarity and dissimilarity degrees for matched and mismatched fragments, respectively.

2.3. VLP-based methods for image-text matching

Although favorable results are obtained in the usage of FE-based and CMA-based methods, they still have limited performance boundaries due to the over-dependency on the visual and textual extractors. With the success of the vision-language large model, recent works [63–65, 68,74] based on vision-language pretraining are trying to transfer the powerful ability of the large model to the downstream task, *e.g.*, image-text matching. According to the differences in the learning modes, present VLP-based methods can be coarsely divided into *finetune-based schemes* and *multimodal large language model-based (MLLM-based) schemes*.

2.3.1. Finetune-based schemes

Approaches following this direction aim to empower the models for the image-text matching task by transferring the strong ability of large language models. Gao et al. propose an accurate and efficient two-stage image-text retrieval model called Contrastive Alignment and visual Contexts modeling (CALiC [64]) for the limitations of the patch-based visual unimodal encoder by applying VLP-based models. In the first stage, a visual unimodal encoder is pre-trained via cross-modal contrastive learning, helping extract easily aligned visual features, and boosting retrieval accuracy and inference speed. In the second stage, a novel visual context is devised to aid the multi-modal encoder in better comprehending visual information for more accurate predictions.

Table 7
Performances of CMA-based Methods on MSCOCO1K dataset.

	Methods	Publications	Text retrieval			Image retrieval			RSum
			R@1	R@5	R@10	R@1	R@5	R@10	
(a) Attention mechanism									
1	SCAN [37]	ECCV2018	72.7	94.8	98.4	58.8	88.4	94.8	507.9
2	CAMP [38]	ICCV2019	72.3	94.8	98.3	58.5	87.9	95.0	506.8
3	RDAN [39]	IJCAI2019	74.6	96.2	98.7	61.6	89.2	94.7	515.0
4	CAAN [40]	CVPR2020	75.5	95.4	98.5	61.3	89.7	95.2	515.6
5	CASC [41]	TNNLS2020	72.3	96.0	99.0	58.9	89.8	96.0	512.0
6	MMCA [42]	CVPR2020	74.8	95.6	97.7	61.6	89.8	95.2	514.7
7	HAN [43]	SIGIR2021	84.3	97.3	99.3	69.7	93.5	97.2	541.3
8	CHAN [44]	CVPR2023	79.7	96.7	98.7	63.8	90.4	95.8	525.1
9	RCAR [45]	TIP2023	80.6	96.6	98.6	64.1	90.5	95.8	526.2
10	LAPS [46]	CVPR2024	78.7	95.5	98.3	66.2	91.3	96.2	526.2
(b) Progressive learning mechanism									
11	IMRAM [47]	CVPR2020	76.7	95.6	98.5	61.7	89.1	95.0	516.6
12	DIME [48]	SIGIR2021	78.8	96.3	98.7	64.8	91.5	96.5	526.6
13	SHAN [49]	IJCAI2021	76.8	96.3	98.7	62.6	89.6	95.8	519.8
14	MEMBER [50]	TIP2021	77.8	96.3	97.9	62.7	89.9	95.1	519.7
15	CMCAN [51]	AAAI2022	81.2	96.8	98.7	65.4	91.0	96.2	529.3
16	HREM [52]	CVPR2023	81.2	96.5	98.9	63.7	90.7	96.0	527.0
17	PPAF [53]	ACMMM2023	80.5	96.8	99.1	66.1	91.9	96.5	530.9
18	IMEB [54]	TCSVT2024	81.0	96.6	98.8	64.1	90.8	95.9	527.2
(c) Reasoning mechanism									
19	GSMN [55]	CVPR2020	78.4	96.4	98.6	63.3	90.1	95.7	522.5
20	CAEMCL [56]	SIGIR2021	78.9	97.5	98.8	65.7	90.2	96.6	527.7
21	SGRAF [57]	AAAI2021	79.6	96.2	98.5	63.2	90.7	96.1	524.3
22	HAT [58]	CVPR2021	82.5	97.2	99.2	84.5	97.4	98.9	559.7
23	NAAF [59]	CVPR2022	80.5	96.5	98.8	64.1	90.7	96.5	527.1
24	CFM [60]	ACMMM2022	78.6	96.1	98.6	62.5	90.4	95.7	521.9
25	DCIN [61]	ACMMM2023	81.4	96.8	99.0	66.1	92.1	96.6	532.0
26	CORA [62]	CVPR2024	81.7	96.7	99.0	66.0	92.0	96.7	532.1

Table 8
Performances of CMA-based methods on MSCOCO5K dataset.

	Methods	Publications	Text retrieval			Image retrieval			RSum
			R@1	R@5	R@10	R@1	R@5	R@10	
(a) Attention mechanism									
1	SCAN [37]	ECCV2018	46.4	77.4	87.2	34.4	63.7	75.7	384.8
2	CAMP [38]	ICCV2019	50.1	82.1	89.7	39.0	68.9	80.2	410.0
3	CAAN [40]	CVPR2020	52.5	83.3	90.9	41.2	70.3	82.9	421.1
4	CASC [41]	TNNLS2020	47.2	78.3	87.4	34.7	64.8	76.8	389.1
5	MMCA [42]	CVPR2020	54.0	82.5	90.7	38.7	69.7	80.8	416.4
6	CHAN [44]	CVPR2023	60.2	85.9	92.4	41.7	71.5	81.7	433.4
7	RCAR [45]	TIP2023	59.6	85.8	92.4	42.5	71.7	81.8	433.8
8	LAPS [46]	CVPR2024	57.5	84.0	90.8	44.5	74.0	83.6	434.4
(b) Progressive learning mechanism									
9	IMRAM [47]	CVPR2020	53.7	83.2	91.0	39.7	69.1	79.8	416.5
10	DIME [48]	SIGIR2021	59.3	85.4	91.9	43.1	73.0	83.1	435.8
11	MEMBER [50]	TIP2021	54.0	81.7	89.6	40.2	70.5	81.1	417.1
12	CMCAN [51]	AAAI2022	61.5	–	92.9	44.0	–	82.6	–
13	PPAF [53]	ACMMM2023	60.0	86.0	92.8	43.7	74.1	84.1	440.7
14	IMEB [54]	TCSVT2024	88.8	99.0	–	76.1	95.5	–	–
(c) Reasoning mechanism									
15	SGRAF [57]	AAAI2021	57.8	–	91.6	41.9	–	81.3	–
16	HAT [58]	CVPR2021	58.5	86.1	92.9	58.6	86.6	92.9	475.6
17	NAAF [59]	CVPR2022	58.9	85.2	92.0	42.5	70.9	81.4	430.9
18	CFM [60]	ACMMM2022	56.8	84.4	91.5	40.2	70.6	81.4	424.9
19	DCIN [61]	ACMMM2023	60.8	86.3	93.0	44.0	74.6	84.3	443.0
20	LEAPRR [76]	SIGIR2023	64.3	87.3	93.1	45.5	74.6	83.7	448.5
21	CORA [62]	CVPR2024	63.0	86.8	92.7	44.2	73.9	84.0	444.6

Xie et al. propose a novel Token Embeddings AlignMent (TEAM [65]) for the problem of implicit supervision on the alignment between visual and textual tokens. The architecture of TEAM is depicted in Fig. 13. Empirically, TEAM explicitly aligns visual and textual tokens, generating token-level matching scores to measure the fine-grained similarity between an input image and text. By following the assumption that the text contains more abstract information, models may be trapped in unsatisfactory performance. To this aim, Dong et al. exhibit

a GTMIS [63] from the perspective of information difference with weak alignment. The core of GTMIS is an imagination network, which can bridge the cross-modal heterogeneity gap and balance information discrepancies. Very recently, Zhang et al. present a Unified Semantic Enhancement momentum contrastive learning (USER [68]) to differentially treat each image region and break through the scale limitation of negative sample pairs. For achievement, two simple yet effective modules are carefully designed, termed Global representation based

Table 9
Performances of CMA-based Methods on Flickr30K Dataset.

	Methods	Publications	Text retrieval			Image retrieval			RSum
			R@1	R@5	R@10	R@1	R@5	R@10	
(a) Attention Mechanism									
1	SCAN [37]	ECCV2018	67.4	90.3	95.8	48.6	77.7	85.2	465.0
2	CAMP [38]	ICCV2019	68.1	89.7	95.2	51.5	77.1	85.3	466.9
3	CAAN [40]	CVPR2020	70.1	91.6	97.2	52.8	79.0	87.9	478.6
4	CASC [41]	TNNLS2020	68.5	90.6	95.9	50.2	78.3	86.3	469.8
5	MMCA [42]	CVPR2020	74.2	92.8	96.4	54.8	81.4	87.8	487.4
6	HAN [43]	SIGIR2021	86.2	96.3	99.0	71.5	90.9	94.9	538.8
7	CHAN [44]	CVPR2023	79.7	94.5	97.3	60.2	85.3	90.7	507.7
8	RCAR [45]	TIP2023	78.7	94.6	97.6	59.5	84.0	89.5	503.9
9	LAPS [46]	CVPR2024	74.0	93.4	97.4	62.5	87.3	92.7	507.3
(b) Progressive Learning Mechanism									
10	IMRAM [47]	CVPR2020	74.1	93.0	96.6	53.9	79.4	87.2	484.2
11	DIME [48]	SIGIR2021	81.0	95.9	98.4	63.6	88.1	93.0	520.0
12	SHAN [49]	IJCAI2021	74.6	93.5	96.9	55.3	81.3	88.4	490.0
13	MEMBER [50]	TIP2021	76.1	93.9	96.7	57.5	83.3	89.9	497.4
14	CMCAN [51]	AAAI2022	79.5	95.6	97.6	60.9	84.3	89.9	510.5
15	HREM [52]	CVPR2023	81.4	96.5	98.5	60.9	85.6	91.3	514.2
16	PPAF [53]	ACMMM2023	83.2	96.6	98.6	64.5	87.9	93.0	523.8
17	IMEB [54]	TCSVT2024	80.0	96.0	98.1	60.0	85.9	91.5	511.5
(c) Reasoning Mechanism									
18	GSMN [55]	CVPR2020	76.4	94.3	97.3	57.4	82.3	89.0	496.7
19	CAEMCL [56]	SIGIR2021	78.7	94.5	97.9	58.2	83.6	89.6	502.5
20	SGRAF [57]	AAAI2021	77.8	94.1	97.4	58.5	83.0	88.8	499.6
21	HAT [58]	CVPR2021	77.0	95.1	97.7	65.4	90.1	92.7	518.0
22	NAAF [59]	CVPR2022	81.9	96.1	98.3	61.0	85.3	90.6	513.2
23	CFM [60]	ACMMM2022	77.6	94.9	97.3	57.5	83.9	90.0	501.2
24	DCIN [61]	ACMMM2023	83.0	96.4	98.6	63.3	87.8	92.4	521.5
25	LEAPRR [76]	SIGIR2023	86.2	96.7	98.9	71.1	89.5	93.2	535.6
26	CORA [62]	CVPR2024	82.3	96.1	98.0	63.0	87.4	92.8	519.6

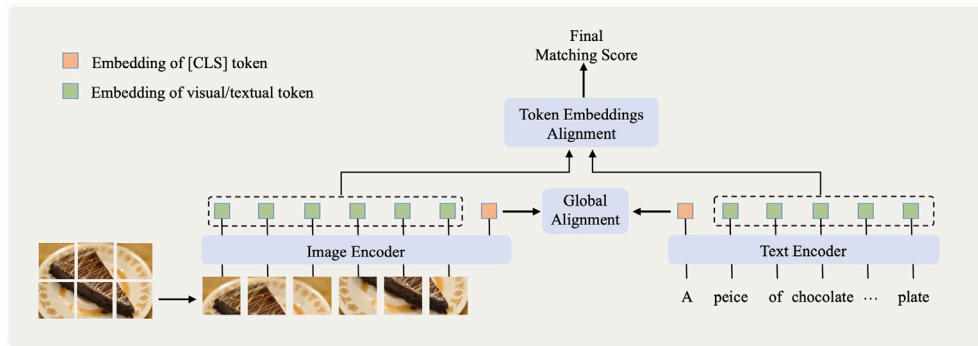


Fig. 13. A representative approach in the VLP-based direction. TEAM [65] transfers the strong ability of large language models for image–text matching.

Semantic Enhancement (GSE) and Self-Guided Enhancement (SGE), to improve global representations cooperatively.

2.3.2. MLLM-based schemes

Rather than merely transferring the ability of large language models, this direction aims to design multiple complementary tasks to improve the retrieval accuracy. In order to relate the visual representation with the appropriate phrases, Lu et al. [69] propose a joint model, ViLBERT, for learning task-agnostic visual grounding from paired vision and linguistic data. The core design is the independent processing streams for visual and linguistic information, which interact via co-attentional transformer layers. This architecture not only addresses the distinct computational requirements of each modality but also enables cross-modal interaction at multiple levels of representational abstraction. Later, Li et al. [70] propose a contrastive loss to align the image and text representations before fusion, namely ALBEF, enabling more grounded vision and language representation learning. The above

approaches mainly rely on noisy image–text pairs collected from the web, which is a suboptimal source of supervision. To resolve this obstacle, Li et al. [72] present a novel BLIP framework that leverages noisy web data through a caption bootstrapping mechanism, where a caption generator produces synthetic captions and a filter module eliminates noisy ones. The architecture of BLIP is depicted in Fig. 14. Concretely, there are two core ingredients within the BLIP: a multimodal mixture of encoder–decoder (MED) and a captioning and filtering (CapFilt), which can operate as a unimodal encoder and learning from noisy image–text pairs, respectively. To avoid the catastrophic forgetting issue raised by previous works, Cui et al. [73] propose a new framework called dynamic knowledge rectification (DKR) that simultaneously achieves incorrect knowledge filtering and rectification. Specifically, DKR starts with filtering the incorrect affinity knowledge computed by the old model on new data. Subsequently, it designs a knowledge rectification method to correct faulty affinities while retaining valid ones. Later, it rectifies new data uniquely retrievable

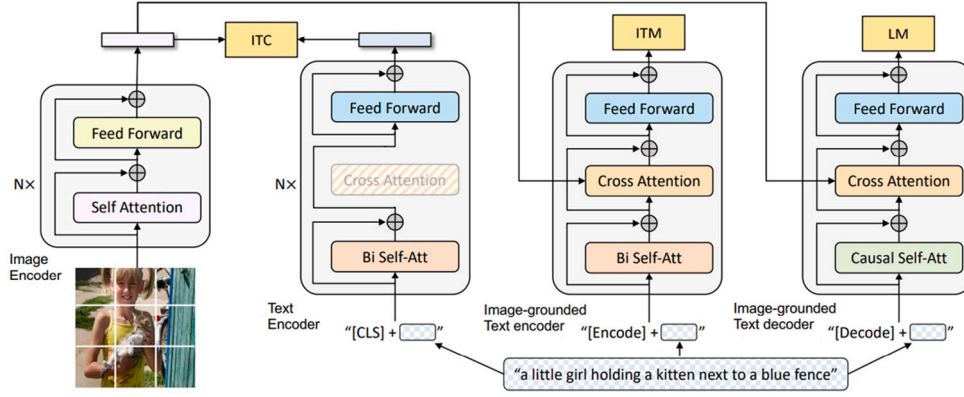


Fig. 14. A representative approach in the VLP-based direction. BLIP [72] designs multiple complementary tasks to improve the retrieval accuracy.

Table 10
Performances of VLP-based methods on MSCOCO5K dataset.

Methods		Publications	Text retrieval			Image retrieval			RSum
			R@1	R@5	R@10	R@1	R@5	R@10	
(a) Finetune-based methods									
1	CALiC [64]	ACMMM2022	79.1	94.4	97.5	61.8	85.1	90.9	508.8
2	TEAM [65]	ACMMM2022	74.9	91.8	95.3	54.7	79.5	86.6	482.8
3	GTMIS [63]	ACMMM2023	81.5	96.0	98.2	65.6	87.8	93.0	522.1
4	USER [68]	TIP2024	67.6	88.4	93.5	47.7	75.1	83.7	456.0
(b) MLLM-based methods									
5	ALBEF [70]	NIPS2021	77.6	94.3	97.2	60.7	84.3	90.5	504.6
6	ViLT [71]	ICML2021	61.5	86.3	92.7	42.7	72.9	83.1	439.2
7	BLIP [72]	ICML2022	82.4	95.4	97.9	65.1	86.3	91.8	518.9
8	DKR [73]	AAAI2024	51.7	75.4	–	29.7	54.2	–	–
9	CUSA [74]	AAAI2024	83.3	96.6	98.5	67.1	87.6	92.7	525.8
10	C-CLIP [75]	ICLR2025	56.9	–	–	42.8	–	–	–

by the new model with corresponding new affinities to prevent negative transfer. Lastly, DKR introduces paired ground-truth labels for data unreachable by either model to facilitate the joint acquisition of old and new knowledge. Alternatively, Liu et al. propose a novel framework named C-CLIP, which not only prevents forgetting but also enhances new task learning impressively by developing continual learning. Facing the inherent problems: inter-modal matching missing and the intra-modal semantic loss. Huang et al. propose a Cross-modal and Uni-modal Soft-label Alignment (CUSA [74]), which has two alignment techniques namely a cross-modal soft-label alignment and a unimodal soft-label alignment to address false negatives and improve similarity recognition. Although the methods above have improved performance on various joint vision-and-language downstream tasks, they are still hard to balance accuracy and efficiency. With this in mind, Kim et al. propose a Vision-and-Language Transformer (ViLT) to handle two modalities in a single unified manner. It is primarily through its shallow, convolution-free pixel-level input embedding. By eliminating deep visual embedders exclusively tailored for image inputs, ViLT achieves a substantial reduction in model size and inference time by design.

3. Performance comparisons

3.1. Datasets and evaluation metrics

Current works for the image-text matching task mainly testify on two benchmark datasets, MSCOCO [77] and Flickr30K [78]. The details are listed in Table 12. Some exhibitions of dataset samples are displayed in Fig. 15.

MSCOCO. It includes 123,287 images, and each image is annotated with 5 annotated captions. MSCOCO is decomposed into 113,287, 5000, and 5000 images for training, validation, and testing. Notably,

retrieval results are tested either by averaging over 5 folds of 1 K test images or on the full 5 K images.

Flickr30K. It contains 31,783 images, and each image is also annotated with 5 captions. Wherein, 29,783 images are chosen as the training set, 1000 images as the validation set, and the rest are the testing set.

R@K and RSum. In this task, the performance of image-text matching is commonly measured by the recall at K (R@K), where the fraction of queries in which the correct item is retrieved in the closest K points to the query. The calculation process is as follows:

$$R@K = \frac{TP@K}{TP@K + FN@K} \quad (1)$$

where TP@K represents the number of closest K items to the query in the top-K results, and FN@k is the number of items that were not in the top-K results. In addition, the RSum values are calculated to comprehensively evaluate the given method, where the RSum values are obtained by summing all of the recall at K for the corresponding method, which can be signified as.

$$RSum = R@1 + R@5 + R@10 \quad (2)$$

3.2. Analysis of quantitative results

To vertically compare the effectiveness of different types of solution ideas and the performance of different methods within each type, the performances of different methods for different types are depicted in Table 4–11 and Fig. 16. By meticulous observation, it can come down to four analyses:

Analyze 1. By observing the results of Tables 4–6 and Fig. 16(a) that resolve the image-text matching task from the perspective of feature enhancement, it can be seen that the methods adopting a novel learning scheme evidently obtain the best performances. This phenomenon

Table 11
Performances of VLP-based methods on Flickr30K dataset.

	Methods	Publications	Text retrieval			Image retrieval			RSum
			R@1	R@5	R@10	R@1	R@5	R@10	
(a) Finetune-based methods									
1	CALiC [64]	ACMMM2022	96.8	100.0	100.0	86.3	97.7	98.9	579.7
2	TEAM [65]	ACMMM2022	90.0	98.0	98.6	71.8	90.5	93.7	542.6
3	GTMIS [63]	ACMMM2023	92.7	99.2	99.7	82.1	97.3	99.2	570.2
4	USER [68]	TIP2024	86.3	97.6	99.4	69.5	91.0	94.4	538.2
(b) MLLM-based methods									
5	ViLBERT [69]	NIPS2019	–	–	–	58.2	84.9	91.5	–
6	ALBEF [70]	NIPS2021	95.9	99.8	100.0	85.6	97.5	98.9	577.7
7	ViLT [71]	ICML2021	83.5	96.7	98.6	64.4	88.7	93.8	525.7
8	BLIP [72]	ICML2022	97.4	99.8	99.9	87.6	97.7	99.0	581.4
9	DKR [73]	AAAI2024	78.5	95.7	–	58.7	83.6	–	–
10	CUSA [74]	AAAI2024	98.5	100.0	100.0	91.3	98.8	99.5	588.1
11	C-CLIP [75]	ICLR2025	84.4	–	–	73.7	–	–	–



Fig. 15. The exhibition of some dataset samples.

Table 12
Summary of benchmark datasets.

Datasets(Image Text)	Training	Validation	Testing	Total
Flickr30K	29783 148915	1000 5000	1000 5000	31783 158915
MSCOCO	113287 566435	5000 25000	5000 25000	123287 616435

demonstrates that the methods adopting the novel learning scheme are superior compared to the other two kinds of methods. The reasons may turn out to be that the methods lying in this direction usually explore appropriate metric functions, effective training mechanisms, or robust fusion skills, which cater to the expectation of the image–text matching task that whether the extracted multimodal features match. However, the other two kinds of methods focus on improving multimodal features, for which the performance of the upper-bound heavily relies on the capability of the backbone architecture to capture the multimodal features. Besides, the performances by the usage of multimodal semantic reinforcement are typically weaker than the methods through single-modal semantic reinforcement, which deviates from common sense. It may result from the reason that the models using multimodal semantic reinforcement increase the model complexity and lack proper cross-modal interaction mechanisms, leading to the difficulty of model learning.

Analyze 2. As depicted in Tables 7–9 and Fig. 16(b), it is not difficult to find that the performances following the reasoning mechanism are considerably better than the other two kinds of methods, manifesting the significance of adopting plausible reasoning ways for the hidden information excavation. Comparing the methods of using the attention mechanism with that of employing the progressive learning mechanism, the latter performances are slightly superior than the

former, where the reason may be analyzed as that repeatedly using the extracted features is considerably necessary, even though the salient information is emphasized with the attention mechanism.

Analyze 3. Observing Tables 10–11 and Fig. 16(c), the approaches following MLLM-based schemes are obviously superior to those of finetune-based schemes. These can come down to two reasons. Firstly, the approaches of MLLM-based schemes usually exploit more powerful large language models. For example, CUSA [74] individually selects Unicom [79] and Sentence-BERT [80] two large models to extract image and text features. Notably, the performance of Unicom outperforms that of the CLIP because Unicom is pre-trained on a large dataset containing a huge amount of data. Secondly, various tasks are designed in the approaches of MLLM-based schemes, which can complement each other. Take the BLIP as an example, the designed image captioning task can implicitly align the image–text semantics, which can benefit the image–text matching task. Based on this analysis, it is important to select powerful large models to improve the retrieval accuracy.

Analyze 4. According to the results from Tables 4–11 and Fig. 16(d), the performances following the vision-language pretraining surpass those using feature enhancement and cross-modal alignment by a large gap, indicating that the selection of backbone architectures is better than the algorithm design. In addition, the performances with the usage of cross-modal alignment slightly exceed those through feature

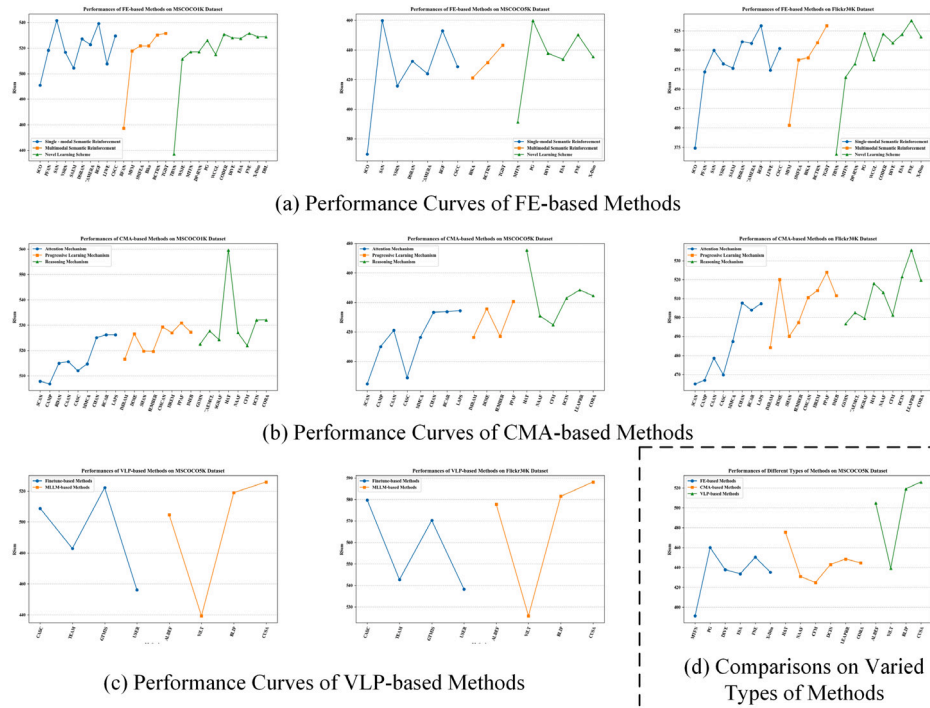


Fig. 16. Performance curves on different types of methods.

Image Retrieval	Text Retrieval
Query: A closeup of the inside of an oven with pizza in it. 	Query:  Top 1: A closeup of the inside of an oven with pizza in it. Top 2: A brick oven with logs and a uncooked pizza next to it. Top 3: There is a stone oven pizza near the fire place.
Query: Several boys on a field playing with a frisbee. 	Query:  Top 1: Several boys on a field playing with a frisbee. Top 2: A bunch of guys that are standing in the grass with a frisbee. Top 3: Young men playing frisbee in a grassy field.
Query: A very big pretty green vase with some flowers. 	Query:  Top 1: A very big pretty green vase with some flowers. Top 2: a vase with some dead vegetation inside of it. Top 3: A beautiful vase full of flowers and pictures next to it.

Fig. 17. Visualization of image-text mutual retrieval on the MSCOCO dataset.

enhancement, implying that it is more significant to eliminate the semantic gaps rather than independently enhance the visual and textual representation ability.

In addition, the mutual retrieval performance is listed according to the image query and the text query in Fig. 17.

4. Conclusions and future works

This work provides a systematic and comprehensive overview of representative methods for the image-text matching task that researchers are increasingly studying. Firstly, most researches focus on enhancing the extracted visual and textual features, therefore, a series of methods are in an attempt to resolve this task, such as single-modal semantic reinforcement, multimodal semantic reinforcement,

novel learning scheme, and so on. Although impressive, they are limited by their performances due to the inadequate interaction between image and text. With this in mind, current researchers shift their attention toward the cross-modal alignment, which satisfies the target of whether the given image-text pair matches. Based on this, present works lying in this direction can be roughly classified into three categories, including attention mechanism, progressive learning mechanism, and reasoning mechanism. However, the performances above are still contained because of the high dependence on ineffective backbones. To this end, recent works tend to introduce prevalent and powerful large models as the new image-text extractors for further improving the effectiveness of the models. Though the performances have improved steadily, current methods for image-text matching still have limited performance. In

the future, the following ideas can be considered to further improve existing methods:

(1) Continue to introduce large models and combine small models to improve cross-modal representation capabilities and enable more intensive alignment. A straightforward idea is to use backbone models with stronger representation capabilities to extract visual and textual features. Due to the extensive knowledge of large models, they can not only be considered backbone networks for extracting features of data in each modality but also can effectively bridge the “heterogeneous gap” in cross-modal data because of the feature alignment capabilities of cross-modal large models.

(2) Introducing wavelet transform to model the global context while preserving the local details of the key objects. The most advanced method currently adopts modeling global contextual information in Fourier space, but the Fourier transform only considers modeling global information without preserving details of local key objects. Simply introducing local details in pixel space may cause feature “drift” problems again due to spatial inconsistency. Wavelet transform can simultaneously model global context and preserve local semantic details in the frequency domain space, which can effectively compensate for the shortcomings of previous methods.

(3) Replacing Transformers with the prevailing Mamba architecture to achieve lightweight global dependency modeling. Thanks to the powerful long-term dependency modeling and feature learning capabilities of Transformer, the state-of-the-art methods have achieved unprecedented performance improvements. However, the excessive computational overhead caused by long-term dependency modeling in the Transformer severely restricts its application in real-world scenarios. Alternatively, the currently popular Mamba mechanism can be used to solve the above challenges in a friendly manner due to its linear global computational cost.

(4) Introduce a more comprehensive modeling of object relationships using knowledge graphs. For modeling object association relationships, current methods either rely on graph inference networks to achieve implicit relationship inference or use pretrained detection models to extract candidate object features for relationship modeling. However, such methods ignore the filtering of existing relationships by exploring existing knowledge. Therefore, introducing knowledge graphs to explicitly improve and filter existing relationships can help further enhance performance.

CRedit authorship contribution statement

Chunlong Dong: Writing – review & editing, Writing – original draft, Methodology. **Caiying Wei:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Caiying Wei reports was provided by Silicon Lake College. Caiying Wei reports a relationship with Silicon Lake College that includes: employment. Caiying Wei has patent pending to Licensee. N/A If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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